

1. 2013Q32

Which of the following descriptions about antibiotics is correct?

- A. Antibiotics can engulf pathogens.
- B. Antibiotics can be produced by fungi.
- C. Antibiotics can bind to specific antigens.
- D. Antibiotics can be produced by lymphocytes.

2. 2015Q33

Hepatitis B is transmitted through

- A. insects.
- B. droplets.
- C. body fluid.
- D. skin contact.

3. 2020Q29

Which of the following combinations of causative agents and ways of transmission of infectious diseases is correct?

	Infectious disease	Causative agent	Way of transmission
A.	cholera	virus	food
B.	cholera	bacterium	body fluid
C.	hepatitis B	virus	body fluid
D.	hepatitis B	bacterium	food

LQ

1. 2012Q2

Dengue Fever is a vector borne disease transmitted by mosquitoes. The table below shows two methods adopted by the Government to break the transmission link of this disease. Complete the table below to show how each method works and comment on its advantage *or* disadvantage.

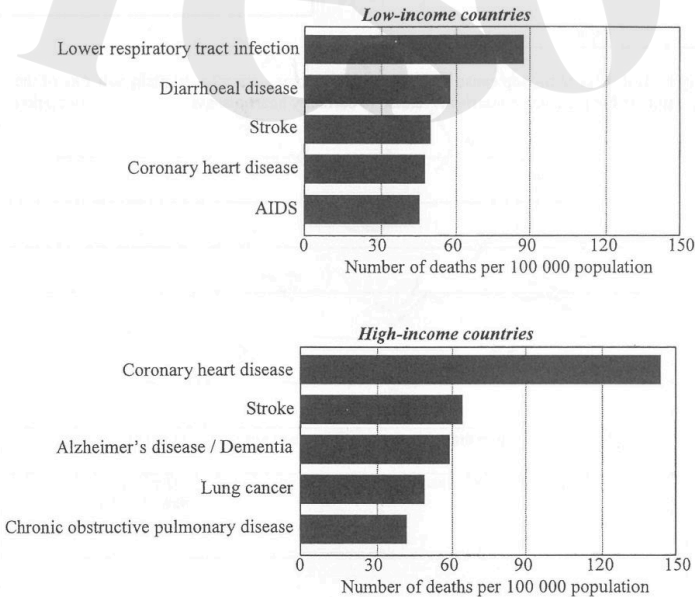
Method	How it works	Comment (advantage or disadvantage)
Spraying of pesticides or larvicidal oil around mosquito’s habitat		
Clearance of accumulated water in a neighbourhood		

2. 2017Q9a

a) What is the vector for transmitting the JE virus? (1 mark)

3. 2018Q4a, b

The bar charts below show the top five diseases that caused death in low-income countries and high income countries respectively in the year 2015:

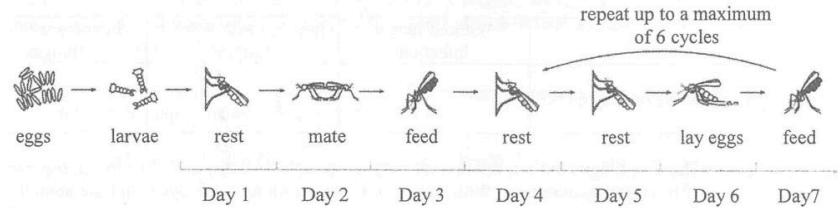


a) With reference to the bar charts, which countries (low-income or high-income) have more infectious diseases as the top five diseases that caused death? (1 mark)

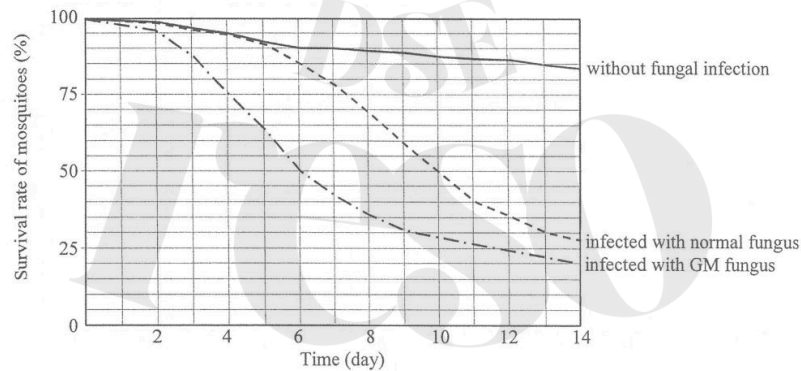
b) Suggest *two* reasons to account for the phenomenon stated in (a). (2 marks)

4. 2020Q7a, c

The diagram below shows the life cycle of mosquitoes:



- a) Give **one** example of a mosquito-borne disease. (1 mark)
- c) To make this fungal species more deadly to mosquitoes, researchers produced a genetically-modified (GM) fungus which can produce a toxin in the mosquitoes. The effectiveness of this biological control was examined by measuring the survival rate of adult mosquitoes over 14 days. The results are shown below:



- i) Using the data from the graph, complete the following table. (1 mark)

	Without fungal infection	Infected with normal fungus	Infected with GM fungus
Survival rate on Day 6 (%)			

- ii) The GM fungus but not the normal fungus was able to ‘wipe-out’ the mosquito population after several generations. With reference to the mosquito’s life cycle and the answer in (ci), explain this phenomenon. (3 marks)

5. 2022Q9a

Antibiotics have been widely used to treat bacterial infections. They work by killing bacteria or inhibiting their growth. However, some strains of bacteria have developed resistance to antibiotics.

- a) How can antibiotics kill bacteria or inhibit their growth? State **three** possible ways. (3 marks)

6. 2023Q4a, b, d

Dengue fever is an infection caused by the dengue viruses (DENV). It is an endemic illness in many countries in tropical and sub-tropical regions. DENV encompasses four different subtypes. Each subtype can lead to dengue fever.

- a) What is the way of transmission for dengue fever? (1 mark)
- b) Suggest **two** environmental factors in tropical and subtropical regions which lead to a higher risk of contracting dengue fever for people living in these regions. Explain your answer. (3 marks)
- d) Suggest **one** preventive measure against the spreading of dengue fever. (1 mark)

Ans

2013Q13: B (47%)

2015Q33: C (80%)

2020Q29: C (70%)

2012Q2

2

Method	How it works	Comment (advantage or disadvantage)	
Spraying of pesticides or larvicidal oil to mosquito's habitat	Directly kills the larvae / adult mosquitoes (1) so that they cannot serve as vector	Pros: quickly puts mosquitoes population down in short term (1) Cons: mosquitoes may develop resistance to the pesticides (1) / environmental contamination when pesticides leak to water bodies / pesticides are toxic to humans	(1),(1)
Clearance of accumulated water in the neighbourhood	Eradicates the breeding places of mosquitoes (1)	Pros: does not have adverse impacts on environment (1) Cons: it is virtually impossible to clear up all stagnant water (1)	(1),(1)

4 marks

2017Q9a

9 (a) • mosquitoes (1)

(1)

2018Q4a, b

4 (a) • low-income countries (1)

(1)

(b) Any **two** of the following:

- poor public / personal hygienic conditions led to easy spreading of infectious diseases (1)
- poor healthcare systems failed to treat patients at critical time (1)
- poor income could not afford the medical costs for treating infectious diseases (1)

(2)

2020Q7a, c

7 (a) dengue fever / malaria / japanese encephalitis / zika / yellow fever (any one, accept other possible answers) (1)

(1)

(c) (i)		Without fungal infection	Infected with normal fungus	Infected with GM fungus	
	Survival rate on Day 6 (%)	90	85	50	(1)

- (ii) • on Day 6, only about 50% of mosquitoes infected with GM fungus survived to reproduce while 85% of adult mosquitoes infected with normal fungus survived to do so (1)
- as a result, the former will have a lower population size than the latter in the next generation (1)
- and the difference would be amplified after several generations (1)

(3)

2022Q9a

9 (a) any **three** of the following:

- prevent formation of bacterial cell wall thereby inhibiting the growth (1)
- break down the bacterial cell membrane, causing lysis to kill the cell (1)
- interfere with the metabolism / protein synthesis thereby inhibiting the bacterial growth (1)
- interfere with the DNA synthesis of the bacteria and stop them from reproduction (1)

(3)

2023Q4a, b, d

4 (a) • it is transmitted by vector / mosquitoes (1)

(1)

(b)	Concept for mark award	
	• correct environmental factor + its effect on mosquitoes (1x2)	(3)
	• overall effect on population (1)	

e.g.

- mosquitoes have a faster growth rate / be more active when the temperature is higher / when it is warmer (1)
- mosquitoes have more breeding grounds during rainy seasons (1)
- therefore, the mosquito population is larger in tropical and subtropical regions (1) resulting in disease spreading more easily

(d) • use insect repellent / wear long-sleeve clothes and trousers / wear clothes with light colours / remove stagnant water bodies (accept other reasonable answers) (1)

(1)